

Chapter 9

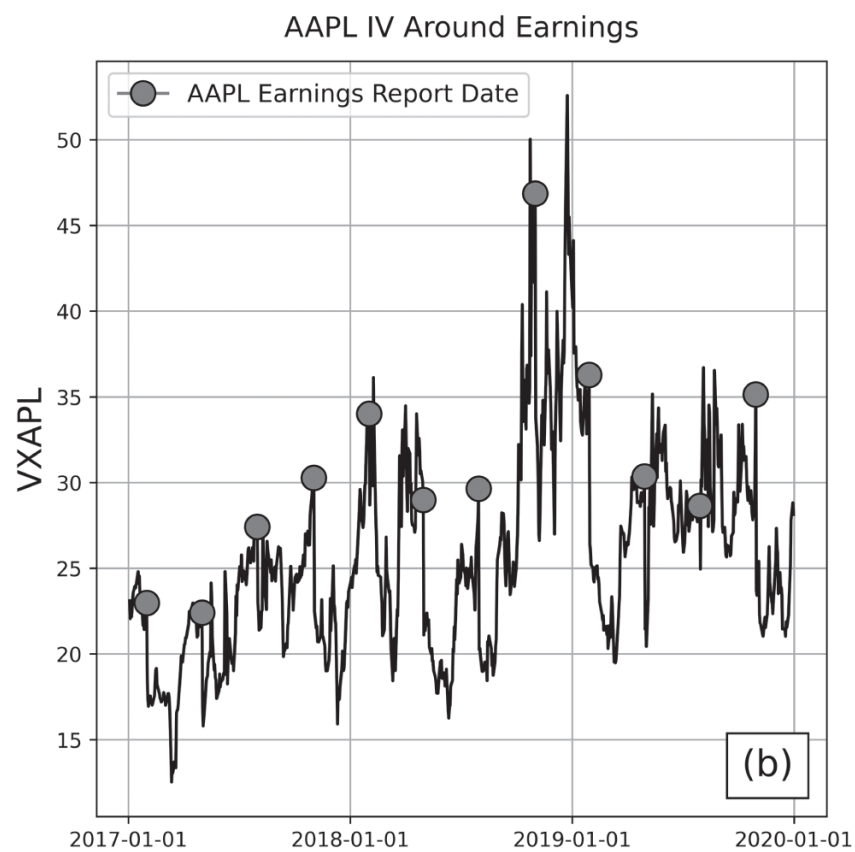
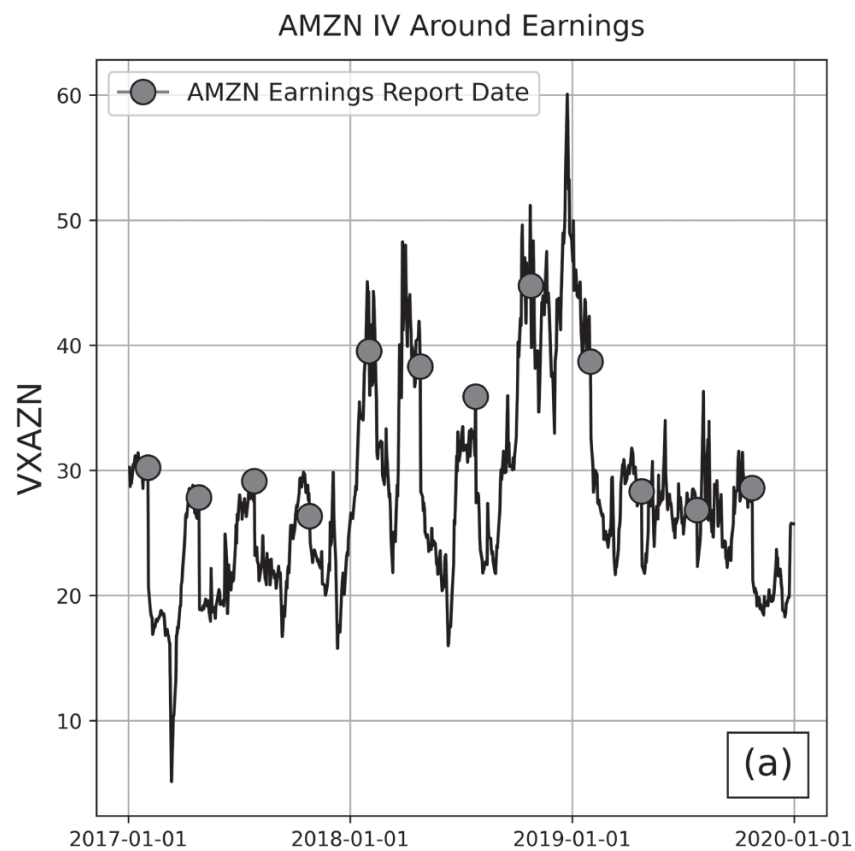
Binary Events

To this point, this book has highlighted unpredictable implied volatility (IV) expansions and their impact on short premium portfolios. However, traders can expect a certain class of IV expansions and contractions with near certainty. These expected volatility dynamics are the result of *binary events*. A binary event is a *known* upcoming event affecting a specific asset (or group of assets) that is *anticipated* to create a large price move. Though price variance is *expected* to increase, it may or may not actually do so depending on the outcome of the binary event.¹ Some examples of binary events include company earnings reports (motivating earnings trades), new product announcements, oil market reports, elections, and Federal Reserve announcements pertaining to the broader market.

Because the date of the anticipated price swing is known, there is typically significant demand for contracts expiring on or after the binary event for that underlying asset. This increased demand results in an increase in the asset's IV, which usually contracts back to nonevent levels immediately after the outcome is known. This trend is shown in [Figure 9.1](#).

The impact from a binary event volatility expansion differs from that of unexpected periods of market volatility because the options approaching binary events are priced to reflect the expectation of large moves in the underlying. However, the high credits and immediate volatility contractions that often result from binary events do not necessarily translate into higher (or even likely) profits for short premium positions. This is because the *magnitude* of the price move following the outcome of the binary event is unpredictable, and it may meet or diverge from expectations. On average, the market response to a binary event tends to be quite large, causing the short options strategies that capitalize on these conditions to be *highly* volatile and not necessarily profitable in the long run. This phenomenon also follows from the efficient market hypothesis (EMH), as the well-understood nature of binary events challenges any consistent edge for these types of strategies.

There is no strong evidence that buying or selling premium around binary events provides a consistent edge with respect to probability of profit (POP) or average profit and loss (P/L) because a lot of the IV overstatement edge is lost in the large moves following a binary event. However, binary event trades are a very time-efficient use of capital because volatility contractions happen more rapidly and predictably than in more regular market conditions. Binary event trades may also be attractive to risk-tolerant traders as a source of market engagement. During earnings season, a single week may present up to 20 high-risk/high-reward opportunities for earnings trades. Binary event trades can also be educational for new traders wanting to learn how to adjust positions in rapidly changing, high volatility conditions outside of sell-offs. These types of trades, as they take place under unique circumstances, are structured and managed differently than typical core or supplemental positions.



[Figure 9.1](#) IV indexes for different stocks from 2017–2020. Assets include (a) AMZN (Amazon stock) and (b) AAPL (Apple stock).

Option Strategies for Binary Events

Because binary event trades are highly volatile and have no strong evidence of a long-term statistical edge, they should only occupy spare portfolio capital and their position size should be kept *exceptionally* small. For example, if a trader's usual position size for an AAPL strangle is a five-lot (five calls and five puts, each written for 100 shares of stock), then an AAPL earnings strangle may comprise a one- or two-lot. Additionally, underlyings for binary event trades are typically stocks, with quarterly earnings reports being the most common type of binary event. Binary event trades take place over much shorter timescales than more typical trades and must be carefully monitored. Earnings trades, for example, are typically opened the day before earnings and closed the day following earnings. This strategy limits downside risk and capitalizes on the majority of the volatility contraction, which tends to occur immediately after the binary event.

The long-term success of binary event trades is difficult to verify because there are relatively few occurrences, resulting in high statistical uncertainty. AAPL, for example, has only reported earnings roughly 100 times since the mid 1990s. The Federal Reserve holds press conferences just eight times per year, and large-scale elections take place once every two or four years. For trading strategies not built around earnings, there are thousands of data points and the statistics are more representative of long-term expectations (the central limit theorem at work). Therefore, working with this small number of data points can yield an *idea* of how binary events trades have performed in the past, but they should be taken with a large grain of salt. [Tables 9.1–9.3](#) demonstrate how earnings trades for three different tech companies have performed over 15 years.

There is clearly significant variability in strategy performance for these three different underlyings. To reiterate, high statistical uncertainty makes it difficult to make definitive conclusions about the success of earnings trades, but some consistent trends are observable. Earnings trades are highly sensitive to changes in time. This is evidenced by the significant differences in the per-trade statistics further from the earnings announcement and

demonstrates why binary event trades must be closely monitored. The *majority* of earnings trades are usually profitable, but do not necessarily average a profit in the long term because of the high per-trade standard deviation. Per-trade variance and tail exposure also tend to increase the longer the trade is held, indicating why these types of trades should be relatively short term. This is why generally, binary event trades, such as earning trades, are closed the day following the binary event.

Table 9.1 Statistics for 45 days to expiration (DTE) 16  AAPL strangles from 2005–2020. Trades are opened the day before an earnings report and closed either one, five, 10, or 20 days after earnings.

	AAPL Strangle Statistics (2005–2020)			
Day Position Is Closed Relative to Earnings	POP	Average P/L	Standard Deviation of P/L	Conditional Value at Risk (CVaR) (5%)
Day After	72%	\$85	\$203	–\$405
5 Days After	70%	\$43	\$400	–\$1,027
10 Days After	61%	\$60	\$408	–\$1,025
20 Days After	56%	–\$34	\$660	–\$1,976

Table 9.2 Statistics for 45 DTE 16 Δ AMZN strangles from 2005–2020. Trades are opened the day before an earnings report and closed either one, five, 10, or 20 days after earnings.

	AMZN Strangle Statistics (2005–2020)			
Day Position Is Closed Relative to Earnings	POP	Average P/L	Standard Deviation of P/L	CVaR (5%)
Day After	65%	\$99	\$803	– \$1,927
5 Days After	65%	\$85	\$842	– \$2,154
10 Days After	72%	\$1	\$1,446	– \$4,416
20 Days After	76%	\$78	\$1,540	– \$4,477

Table 9.3 Statistics for 45 DTE 16 Δ GOOGL strangles from 2005–2020. Trades are opened the day before an earnings report and closed either one, five, 10, or 20 days after earnings.

	GOOGL Strangle Statistics (2005–2020)			
Day Position Is Closed Relative to Earnings	POP	Average P/L	Standard Deviation of P/L	CVaR (5%)
Day After	75%	–\$60	\$1,320	– \$4,639
5 Days After	67%	–\$113	\$1,358	– \$4,724
10 Days After	65%	–\$122	\$1,275	– \$3,675
20 Days After	71%	–\$2	\$1,584	– \$4,909

Takeaways

1. A binary event is a known upcoming event affecting a specific asset (or group of assets) that is anticipated to create a large price move. This anticipation creates demand for options contracts expiring on or after the binary event and an increase in the IV of the asset. IV typically contracts back to nonevent levels immediately after the outcome is known.
2. The high credits and immediate volatility contractions resulting from binary events do not necessarily translate to large or consistent short premium profits because the magnitude of the market response is unpredictable. Binary events trades are generally highly volatile and undependable sources of profit but can be used for market engagement or an educational experience for new traders.
3. Binary event trades should only occupy spare portfolio capital and their position size should be kept *exceptionally* small. Binary event trades should also take place over much shorter timescales than more typical trades, and they must be carefully monitored.

Note

- 1 The term binary is used to describe systems that can exist in one of two possible states (on/off, yes/no). In this context, a binary event is a type of event where price changes either remain within expectations or exceed expectations.